

TP5 — Resumen: Índices, Vistas y Vistas Materializadas

Base de Datos II · Unidad 3 · Semana 5 · Food Store masivo · PostgreSQL 18 · Esquema R7 · OpenCode

Parte A · Plan de indexado

Índice	Antes	Después	Ganancia
idx_producto_nombre_vig (parcial activo, text_pattern_ops)	Seq Scan · 21,5 ms	Bitmap Index Scan · 8,4 ms	Seq→Bitmap (prefijo indexado)
idx_cliente_apellido (B-tree)	Seq Scan · 15,9 ms	Index Scan · 0,9 ms	~18×
idx_pedido_forma_fecha (compuesto)	Bitmap idx_pedido_fecha + Filter (10.375 rem.) · 78,4 ms	Bitmap idx_pedido_forma_fecha · 6,8 ms	sin remoción de filas

Las capturas siguientes son la salida real de EXPLAIN ANALYZE en PostgreSQL 18.

Captura 1

```
explain analyze · producto · ANTES (seq scan)

food_store_tp3=>
SET
SET

                                QUERY PLAN
-----
Seq Scan on producto  (cost=0.00..1434.00 rows=11111 width=32) (actual time=0.033..20.406 rows=11111.00 loops=1)
  Filter: (activo AND ((nombre)::text ~~ 'Producto 1% '::text))
  Rows Removed by Filter: 38889
  Buffers: shared hit=809
Planning:
  Buffers: shared hit=176
Planning Time: 5.669 ms
Execution Time: 21.535 ms
(8 filas)

RESET
RESET
```

Captura 2

```
explain analyze · producto · DESPUÉS (bitmap index

food_store_tp3=>

                                QUERY PLAN
-----
Bitmap Heap Scan on producto  (cost=296.88..1248.14 rows=11111 width=32) (actual time=2.524..7.408 rows=11111.00
loops=1)
  Recheck Cond: activo
  Filter: ((nombre)::text ~~ 'Producto 1% '::text)
  Heap Blocks: exact=185
  Buffers: shared hit=229
  -> Bitmap Index Scan on idx_producto_nombre_vig  (cost=0.00..294.10 rows=11381 width=0) (actual time=2.230..2.230
rows=11111.00 loops=1)
    Index Cond: (((nombre)::text >= 'Producto 1'::text) AND ((nombre)::text <~ 'Producto 2'::text))
    Index Searches: 1
    Buffers: shared hit=44
Planning:
  Buffers: shared hit=12
Planning Time: 0.762 ms
Execution Time: 8.373 ms
(13 filas)
```

Captura 3

```

explain analyze · cliente · ANTES (seq scan)

food_store_tp3=>
SET
SET

                                QUERY PLAN
-----
Seq Scan on cliente  (cost=0.00..517.00 rows=1 width=58) (actual time=0.090..7.075 rows=1.00 loops=1)
  Filter: ((apellido)::text = 'Apellido470'::text)
  Rows Removed by Filter: 19999
  Buffers: shared hit=267
Planning:
  Buffers: shared hit=69
Planning Time: 8.692 ms
Execution Time: 7.100 ms
(8 filas)

RESET
RESET

```

Captura 4

```

explain analyze · cliente · DESPUÉS (index scan)

food_store_tp3=>

                                QUERY PLAN
-----
Index Scan using idx_cliente_apellido on cliente  (cost=0.29..8.30 rows=1 width=58) (actual time=0.114..0.116 rows=1.00
loops=1)
  Index Cond: ((apellido)::text = 'Apellido470'::text)
  Index Searches: 1
  Buffers: shared hit=3
Planning Time: 0.186 ms
Execution Time: 0.281 ms
(6 filas)

```

Captura 5

```

explain analyze · pedido · ANTES (solo idx_pedido_fecha, con Filter)

food_store_tp3=>
DROP INDEX

                                QUERY PLAN
-----
Bitmap Heap Scan on pedido  (cost=334.02..2279.51 rows=5278 width=28) (actual time=1.532..18.500 rows=5140.00 loops=1)
  Recheck Cond: ((fecha >= '2026-01-01 00:00:00-03'::timestamp with time zone) AND (fecha < '2026-02-01 00:00:00-
03'::timestamp with time zone))
  Filter: (forma_pago = 'TARJETA'::forma_pago_enum)
  Rows Removed by Filter: 10375
  Heap Blocks: exact=1667
  Buffers: shared hit=1713
  -> Bitmap Index Scan on idx_pedido_fecha  (cost=0.00..332.70 rows=15628 width=0) (actual time=1.257..1.258
rows=15515.00 loops=1)
    Index Cond: ((fecha >= '2026-01-01 00:00:00-03'::timestamp with time zone) AND (fecha < '2026-02-01 00:00:00-
03'::timestamp with time zone))
    Index Searches: 1
    Buffers: shared hit=46
Planning:
  Buffers: shared hit=83
Planning Time: 5.983 ms
Execution Time: 18.780 ms
(14 filas)

```

Captura 6

explain analyze · pedido · DESPUÉS (idx_pedido_forma_fecha)

food_store_tp3=>
CREATE INDEX

QUERY PLAN

Bitmap Heap Scan on pedido (cost=151.71..1916.08 rows=5278 width=28) (actual time=2.708..4.526 rows=5140.00 loops=1)
 Recheck Cond: ((forma_pago = 'TARJETA'::forma_pago_enum) AND (fecha >= '2026-01-01 00:00:00-03'::timestamp with time zone) AND (fecha < '2026-02-01 00:00:00-03'::timestamp with time zone))
 Heap Blocks: exact=1591
 Buffers: shared hit=1591 read=22
 -> Bitmap Index Scan on idx_pedido_forma_fecha (cost=0.00..150.40 rows=5278 width=0) (actual time=2.494..2.494 rows=5140.00 loops=1)
 Index Cond: ((forma_pago = 'TARJETA'::forma_pago_enum) AND (fecha >= '2026-01-01 00:00:00-03'::timestamp with time zone) AND (fecha < '2026-02-01 00:00:00-03'::timestamp with time zone))
 Index Searches: 1
 Buffers: shared read=22
Planning:
 Buffers: shared hit=22 read=1
 Planning Time: 5.440 ms
 Execution Time: 4.883 ms
(12 filas)

Captura 7

explain · producto por categoria → ya usa idx_producto_categoria_activo (por eso se descartó el índice propuesto)

food_store_tp3=>

QUERY PLAN

Bitmap Heap Scan on producto pr (cost=140.97..1105.91 rows=12475 width=28)
 Recheck Cond: (categoria_id = 1)
 -> Bitmap Index Scan on idx_producto_categoria_activo (cost=0.00..137.85 rows=12475 width=0)
 Index Cond: (categoria_id = 1)
(4 filas)

Costo de escritura: batch de 300 INSERT en pedido (transacción con ROLLBACK).

Estado	Tiempo (300 INSERT en pedido)
Sin los índices nuevos	47,5 ms
Con los 3 índices nuevos	151,6 ms

Parte B · Vistas y verificación de equivalencia

Vista	Filas (vista)	Filas (manual)	Diferencia (EXCEPT)
vista_productos_vigentes (catálogo vigente + categoría)	50.000	50.000	0
vista_pedidos_cliente (pedidos + cliente)	200.000	200.000	0
vista_detalle_pedido (líneas + nombre de producto)	600.000	600.000	0

Captura 8

verificación 1 · (vista) EXCEPT (manual) = 0

food_store_tp3=>
diff

0
(1 fila)

Captura 9

```
verificación 2 · (vista) EXCEPT (manual) = 0

food_store_tp3=>
diff
-----
0
(1 fila)
```

Captura 10

```
verificación 3 · (vista) EXCEPT (manual) = 0

food_store_tp3=>
diff
-----
0
(1 fila)
```

Seguridad: vista_pedidos_cliente no expone cliente.telefono; vista_productos_vigentes no expone descripcion/activo.

Parte C · Vista materializada de facturación

Métrica	Consulta base (4 tablas)	Vista materializada
Tiempo	2447,3 ms	~15 ms
Plan	Nested Loop x3 + GroupAggregate sobre 600.000 líneas	Scan de 60 filas con índice único (mes, categoria)

Captura 11

```
select contra la MV · 60 filas de agregado

food_store_tp3=>
  mes | categoria | facturado
-----+-----+-----
2025-07 | Pizzas    | 16412928.69
2025-07 | Bebidas   | 16321435.87
2025-07 | Postres   | 15796284.91
2025-07 | Empanadas | 15333740.47
2025-08 | Pizzas    | 158540643.64
(5 filas)
```

explain analyze · la misma consulta SIN materializar

food_store_tp3=>

QUERY PLAN

```

-----
Limit (cost=15772.19..15774.12 rows=5 width=72) (actual time=1933.632..1992.329 rows=5.00 loops=1)
  Buffers: shared hit=229528, temp read=605 written=836
  -> Incremental Sort (cost=15772.19..248015.47 rows=600000 width=72) (actual time=1933.629..1992.324 rows=5.00
loops=1)
    Sort Key: (to_char(date_trunc('month'::text, p.fecha), 'YYYY-MM'::text)), (sum(dp.subtotal)) DESC
    Presorted Key: (to_char(date_trunc('month'::text, p.fecha), 'YYYY-MM'::text))
    Full-sort Groups: 1 Sort Method: quicksort Average Memory: 25kB Peak Memory: 25kB
    Buffers: shared hit=229528, temp read=605 written=836
    -> GroupAggregate (cost=15771.09..231760.58 rows=600000 width=72) (actual time=420.516..1992.242 rows=9.00
loops=1)
      Group Key: (to_char(date_trunc('month'::text, p.fecha), 'YYYY-MM'::text)), c.nombre
      Buffers: shared hit=229528, temp read=605 written=836
      -> Incremental Sort (cost=15771.09..216760.58 rows=600000 width=48) (actual time=419.866..1966.188
rows=61855.00 loops=1)
        Sort Key: (to_char(date_trunc('month'::text, p.fecha), 'YYYY-MM'::text)), c.nombre
        Presorted Key: (to_char(date_trunc('month'::text, p.fecha), 'YYYY-MM'::text))
        Full-sort Groups: 3 Sort Method: quicksort Average Memory: 27kB Peak Memory: 27kB
        Pre-sorted Groups: 3 Sort Method: quicksort Average Memory: 2409kB Peak Memory: 3415kB
        Buffers: shared hit=229528, temp read=605 written=836
        -> Nested Loop (cost=15770.14..200505.69 rows=600000 width=48) (actual time=317.745..1827.216
rows=95932.00 loops=1)
          Buffers: shared hit=229528, temp read=605 written=836
          -> Nested Loop (cost=15770.00..183854.97 rows=600000 width=24) (actual
time=317.702..1418.735 rows=95932.00 loops=1)
            Buffers: shared hit=229520, temp read=605 written=836
            -> Nested Loop (cost=15769.70..153327.39 rows=600000 width=24) (actual
time=316.794..844.672 rows=95932.00 loops=1)
              Buffers: shared hit=97979, temp read=605 written=836
              -> Gather Merge (cost=15769.27..38563.39 rows=200000 width=16) (actual
time=316.732..440.275 rows=31978.00 loops=1)
                Workers Planned: 1
                Workers Launched: 1
                Buffers: shared hit=1710, temp read=605 written=836
                -> Sort (cost=14769.26..15063.38 rows=117647 width=16) (actual
time=229.055..244.813 rows=16512.50 loops=2)
                  Sort Key: (to_char(date_trunc('month'::text, p.fecha), 'YYYY-
MM'::text))
                  Sort Method: external merge Disk: 4272kB
                  Buffers: shared hit=1710, temp read=605 written=836
                  Worker 0: Sort Method: external merge Disk: 2400kB
                  -> Parallel Seq Scan on pedido p (cost=0.00..2848.47 rows=117647
width=16) (actual time=0.518..81.507 rows=100000.00 loops=2)
                    Buffers: shared hit=1672
                    -> Index Only Scan using idx_detalle_cover_pedido_prod on detalle_pedido dp
(cost=0.42..0.54 rows=3 width=24) (actual time=0.008..0.011 rows=3.00 loops=31978)
                      Index Cond: (pedido_id = p.id)
                      Heap Fetches: 0
                      Index Searches: 31978
                      Buffers: shared hit=96269
                      -> Memoize (cost=0.30..0.32 rows=1 width=16) (actual time=0.005..0.005 rows=1.00
loops=95932)
                        Cache Key: dp.producto_id
                        Cache Mode: logical
                        Hits: 52085 Misses: 43847 Evictions: 0 Overflows: 0 Memory Usage: 5139kB
                        Buffers: shared hit=131541
                        -> Index Scan using producto_pkey on producto pr (cost=0.29..0.31 rows=1
width=16) (actual time=0.007..0.007 rows=1.00 loops=43847)
                          Index Cond: (id = dp.producto_id)
                          Index Searches: 43847
                          Buffers: shared hit=131541
                          -> Memoize (cost=0.14..0.16 rows=1 width=16) (actual time=0.001..0.001 rows=1.00
loops=95932)
                            Cache Key: pr.categoria_id
                            Cache Mode: logical
                            Hits: 95928 Misses: 4 Evictions: 0 Overflows: 0 Memory Usage: 1kB

```

```
Buffers: shared hit=8
-> Index Scan using categoria_pkey on categoria c (cost=0.13..0.15 rows=1 width=16)
(actual time=0.005..0.005 rows=1.00 loops=4)
    Index Cond: (id = pr.categoria_id)
    Index Searches: 4
    Buffers: shared hit=8
```

Captura 13

```
refresh materialized view concurrently

food_store_tp3=>
REFRESH MATERIALIZED VIEW
===FIN===
```

Frecuencia recomendada: mensual (corte contable). Implicancia: entre refrescos la MV queda congelada al último estado — aceptable para corte mensual; para datos del día, refrescar on-demand o usar la consulta base.

DUIA · Declaración de Uso de IA (TP5)

Ítem	Detalle
Herramientas	OpenCode (IA) + PostgreSQL 18 (verificación real con EXPLAIN ANALYZE)
Kiro	No está instalado: specs con la plantilla Kiro redactados a mano
Descarte IA	idx_producto_categoria rechazado: redundante con idx_producto_categoria_activo (Semana 3)
Traducción R7	usuario→cliente · estado/eliminado→activo · idx_pedido_usuario→idx_pedido_cliente
Archivos	indices.sql · views.sql · informe_mediciones.md · duia.md · README.md · specs/